

Abstract Algebra

: Lecture 17

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2. let $E = F(\alpha)$.

★ **Exercise 1.** Let F be a field, and $f(x) \in F[x]$, irreducible. Then there exists an extension E of F s.t. $f(x)$ has a root in E .

证明. Let $E = F[x]/(f(x))$. Since F is a field, $F[x]$ is a PID. $f(x)$ is irreducible shows $(f(x))$ is a maximal ideal hence E is a field. So $\bar{x}$ is a root of $f(x)$ in E since $f(\bar{x}) = \overline{f(x)} = 0$. $\checkmark$. $\square$

Definition 2. F is called a algebraically closed field if any polynomial $f(x) \in F[x]$ is reducible unless $\deg f = 1$.

Definition 3. Let E/F be a field extension. Then E can be view as a vector space over F . If $\dim_F E = n$ is finite, then E is called a finite extension of F of degree n .

Lemma 4. If K/E is of degree m , E/F is of degree n , then K/F is of degree mn . $\checkmark$.

Construction by ruler(straightedge) and compasses.

Given a unit 1.

- (1). We can construct all integers.
- (2). We can construct all rational numbers.
- (3). We can construct all roots of quadratic polynomials.

Now Let $F_0 = \mathbb{Q}$, $F_{n+1} = F_n(\sqrt{a_n})$ where a_n is a square-free integer. Then $[F_{n+1} : F_n]$ equal to 1 or 2.
i.e. if α is constructible, then F_{n+1} is a finite extension of F_0 of degree of 2^k where $k \in \mathbb{Z}_{\geq 0}$.

Let F be a finite field.

(1). Then $|F| = p^d$ where p is a prime number and $d \in \mathbb{Z}_{\geq 1}$.

(2). $(F, +) \simeq \mathbb{Z}_p^d$.

(3). $(F^\times, *) \simeq \mathbb{Z}_{p^d-1}$.

证明. (of (3)): Let m be the exponent of $F^\times$. (m is the least common multiple of the order of elements of $F^\times$) Then each element t of $F^\times$ has order dividing m . So t is a root of $x^m - 1$. Let $a \in F^\times$ with $|a| = m$. Then $m = |a| \mid |F^\times|$ by Lagrange's theorem. So $m \leq p^d - 1$. On the other hand, $x^m - 1$ has at most m roots in F . So $m \geq p^d - 1$. So $m = p^d - 1$. And $F^\times \simeq \mathbb{Z}_{p^d-1}$. $\checkmark$. $\square$

Since F finite.

and F 's els are all roots of $x^m - 1$.

eg. $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$
dim: 2, 4, 8
[$\mathbb{Q}(\sqrt{2}) : \mathbb{Q}$] = 2, [$\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})$] = 2, [$\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) : \mathbb{Q}(\sqrt{2}, \sqrt{3})$] = 2
i.e. $\forall x \in F^\times, x^m = 1$, and m minimum.

Consider the group action of multiple group on addition group, we can get a semidirect product of two groups such as $Z_p^d : Z_{p^d-1}$, denoted by $\text{AGL}_1(p^d)$.

Let F be a finite field of order p^d , i.e. $\mathbb{F}_{p^d}$ or denoted by $\text{GF}(p^d)$.

Theorem 5. $\phi : F \rightarrow F$ s.t. $x \mapsto x^p$ is an automorphism of F . $F : \text{GF}(p^d)$. $\text{char} = p$.

证明. Check: $(xy)^\phi = x^p y^p = x^\phi y^\phi$, $(x+y)^\phi = x^\phi + y^\phi$. This is called Frobenius automorphism. $\square$

Theorem 6. Let F be a field of characteristic 0, then a finite extension of F is a simple extension.

证明. Let $E = F(\alpha, \beta)$, Let $f(x), g(x)$ be irreducible polynomials in $F[x]$ s.t. $f(\alpha) = 0, g(\beta) = 0$. Let $\gamma = \alpha + c\beta$ where $c \in F$. We need to determine c s.t. $F(\alpha, \beta) = F(\gamma)$. $\checkmark$.
 Let $h(x) = f(\gamma - cx) \in F(\gamma)[x]$, then $h(\beta) = f(\alpha) = 0$. So β is root of $h(x)$ and $g(x)$. If β is the only common root of $h(x)$ and $g(x)$, then $x - \beta = \gcd(g(x), h(x)) = s(x)g(x) + t(x)h(x) \in F(\gamma)[x]$.
 So $\beta \in F(\gamma)$, and $\alpha = \gamma - c\beta \in F(\gamma)$. So $F(\alpha, \beta) = F(\gamma)$. $\checkmark$.

We will finish the proof next time. $\square$